

Project Initialization and Planning Phase

Date	12 August, 2026
Team ID	Individual
Project Name	AI Knowledge Retention Predictor
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report aims to transform loan approval using machine learning, boosting efficiency and accuracy. It tackles system inefficiencies, promising better operations, reduced risks, and happier customers. Key features include a machine learning-based credit model and real-time decision-making.

Project Overview	
Objective	Implement advanced machine learning and AI techniques to model individual memory decay, enabling faster, personalized knowledge retention tracking and automated quiz generation.
Scope	Provide an end-to-end web platform using Streamlit that tracks user recall, calculates mathematical forgetting curves, and dynamically prompts revision schedules.
Problem Statement	
Description	Traditional study methods rely on inefficient, static revision schedules and last-minute cramming, leading to rapid cognitive memory loss and unmonitored skill decay.
Impact	Untracked learning retention results in student anxiety, low exam performance, and poor long-term subject mastery.

Proposed Solution	
Approach	Combining Ebbinghaus forgetting curve algorithms with LLM-powered adaptive quizzes to measure and predict individual topic decay in real time.
Key Features	• Implementation of a mathematical memory retention prediction model. • Real-time generation of contextual quizzes and hints via Google Gemini API • Continuous adaptation based on historical quiz performance logs.

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing	CPU/GPU specifications and execution power	Standard Cloud CPU / Basic GPU
Memory	RAM allocation	8 GB RAM
Storage	Disk space for data, SQLite database, and logs	10 GB Cloud SSD
Software		
Frameworks	Python web application frameworks	Streamlit
Libraries	Data processing, plotting, and math libraries	NumPy, SciPy, Pandas, Plotly
Development Environment	IDE / Code Editors	VS Code, Jupyter Notebook
Data		
Data	Database formats and API inputs	SQLite Database (user_history.db), Google Gemini API